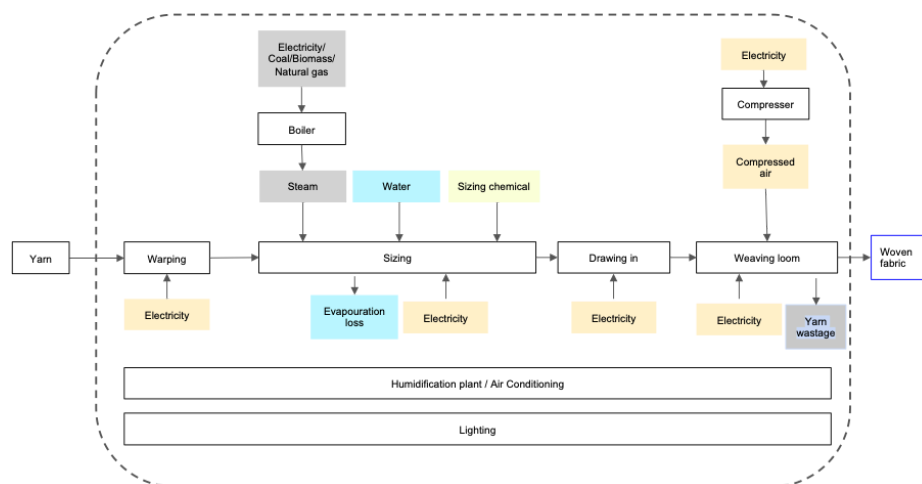




Weaving

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▼ 1. Introduction



The life cycle inventory (LCI) review and modelling presented in this study were undertaken to develop a transparent, literature-supported, and verifiable representation of weaving processes. The approach is designed to provide traceable inventory data, clear methodological documentation, and a structure that can be updated as new evidence becomes available.

1.1 Purpose of the study

The purpose of this study is to develop fully traceable, transparent, and updatable life cycle inventory (LCI) datasets for 1 kg of woven fabric, covering the most relevant process variants currently represented in the Environmental Footprint (EF) 3.1 dataset library. The aim is to replace the existing EF datasets with LCIs that are scientifically robust, verifiable, and aligned with real-world textile production technologies.

This work involves:

- Conducting a detailed literature review of textile weaving, preparation, and ancillary processes
- Collecting secondary data from peer-reviewed studies, industrial benchmarks, and sector reports

1.2 Weaving process

Weaving is the process of interlacing two sets of yarns (warp and weft) to form a fabric. It is one of the oldest and most fundamental textile manufacturing operations. Globally, woven fabrics accounted for approximately 47.5% of the textile fabric market in 2024 ([Mordor Intelligence report](#)). Because weaving remains a large-scale operation in textile mills and

involves high-speed machines, compressed air, electricity, and auxiliary systems, it is an energy-intensive process and therefore critical to investigate for footprint modelling, efficiency improvement, and decarbonisation.

1.3 System boundary

Figure 1. System boundary of the weaving LCI model.

- **Warping** – Combines yarns from multiple cones into a parallel sheet and winds them onto a warp beam for weaving.
- **Sizing** – Applies a water-based protective coating to warp yarns to increase strength and reduce breakage during weaving.
- **Drawing-in / knotting** – Threads each warp yarn through heddles and reed, or ties yarn ends, to prepare the loom for the selected fabric pattern.
- **Weaving (loom operation)** – Interlaces warp and weft yarns using mechanisms such as rapier, air-jet, projectile, or shuttle insertion to form fabric.

1.4 Functional unit and scope

Functional unit: 1 kg of woven fabric ready for wet processing

Inclusions:

- Direct and indirect energy consumption (electricity and thermal energy) associated with weaving processes
- Sizing chemicals
- Lubricating oil
- Water consumption and losses due to evaporation
- Capital goods
- Auxiliary waste flows (waste mineral oil, waste paperboard, end-of-life machinery)

Exclusions:

- Packaging and distribution
- Yarn production and its associated upstream process stages (fibre production)
- Infrastructure-related impacts
- **Yarn losses** — modelled externally via a loom-specific input/output ratio (IOR), rather than inside the weaving dataset. The caller applies the IOR both to scale the upstream material input and to book the end-of-life treatment of the waste yarn (see §3.2.3)

▼ 2. Data sources and literature review

The objective of this literature review was to identify the weaving process variants represented in existing LCIs that require reconstruction using transparent and traceable data, assess the gaps and limitations in current LCI datasets available in databases such as EF 3.1 and ecoinvent, and analyse the key factors that influence the life cycle inventory of weaving processes.

2.1 Weaving datasets in the EF 3.1 database

Observations

- EF 3.1 includes weaving-as-a-service datasets divided by loom type (shuttle, rapier, projectile, Jacquard, air-jet, water-jet, and generic “other” looms), fabric type (cotton, twill cotton, polyester, carpet, silk, and fabrics made with spin-dyed yarn), and yarn linear densities (45–370 dtex, including an average-dtex variant).
- The modelling implicitly assumes that when one parameter (for example loom type or dtex) is varied, all other parameters remain constant, although this is not explicitly documented.

- Electricity demand in the EF 3.1 weaving datasets is derived almost entirely from [Koç & Cincik \(2010\)](#), which provides reference electricity consumption for a cotton fabric woven with **Ne 30 (~196 dtex)**. EF modelling then applies a scaling factor based on the assumption that finer yarns require more picks per unit area and therefore more electrical energy per kilogram. The normalization reference (196 dtex) is maintained across all variants, regardless of fabric type.
- Differences in energy consumption between loom types (air-jet, shuttle, rapier, projectile, and water-jet) are adjusted using ratios reported in [Prasanna \(2016\)](#). However, the source cited by Prasanna et al. for these ratios is not traceable, and the underlying data behind the ratios are unclear, which reduces confidence in the adjustment factors.
- Lubricant consumption is assumed or drawn from [Roos \(2019\)](#), without confirmation of technology age or regional representativeness.
- EF 3.1 documentation does not clearly state whether **sizing** is included within the system boundaries for weaving.
- Although EF 3.1 contains fabric variants (cotton, polyester, carpet, and silk), the underlying [Koç dataset](#) is based **only on cotton**, and no transparent explanation is provided for how parameters such as warp density, fabric GSM, or loom efficiency were adapted for different materials. The methodology used to transfer cotton-specific data to non-cotton fabrics is not available.

Based on the EF 3.1 dataset review, a total of **22 weaving variants** can be reconstructed with transparent and traceable LCIs.

Loom type	Material	Yarn dtex
Shuttle loom	Cotton fabric	45 dtex
Rapier loom	Twill cotton fabric	70 dtex
Projectile loom	Polyester fabric	120 dtex
Air-jet loom	Silk fabric	150 dtex
Water-jet loom	Carpet fabric	170 dtex
Jacquard loom	Fabric woven from spin-dyed yarn	200 dtex
		250 dtex
		300 dtex
		330 dtex
		370 dtex

2.2 Observations from the literature review of peer-reviewed studies on weaving

1. **Very limited availability of full LCIs for weaving.** Only a handful of studies provide an end-to-end life cycle inventory for weaving, with [Roos et al.](#) serving as the primary comprehensive reference. Most other publications report only partial data or focus on isolated process parameters.
2. A majority of weaving-related research concentrates on **electrical energy consumption at the loom**, while **thermal energy from sizing** is either excluded or insufficiently detailed. This results in incomplete modelling of weaving preparation stages.
3. Most journal articles do not report essential parameters such as **yarn linear density (dtex/tex/denier)** or **loom type**. These are critical drivers of energy consumption and breakage behaviour, and their absence prevents LCIs from being scaled or compared reliably across processes.
4. Numerous studies rely heavily on [Koç \(2010\)](#) for both electrical and thermal energy estimates, even though:
 - a. The data is more than a decade old.
 - b. The modelled mill is **air-jet-dominated**, which inherently requires higher compressed-air consumption.

- c. The reported **thermal energy for sizing (9.85 kJ/kg)** appears implausibly low, suggesting either a unit error or a misinterpretation, as realistic thermal demand is significantly higher.

This reliance introduces systematic uncertainty into subsequent LCIs.

5. Databases such as **Idemat** and studies such as **Van der Velden et al.** use interpolation or extrapolation models to scale weaving energy based on yarn dtex, typically assuming inverse proportionality. However, empirical validation of these models remains limited.
6. There is **no peer-reviewed LCA** that systematically compares environmental impacts across **different weaving technologies** (air-jet, rapier, projectile, shuttle, and water-jet). EF 3.1 attempts this by applying “energy consumption ratios,” but these ratios originate from **non-transparent or untraceable sources**, rather than from rigorous LCI studies.
7. Most weaving-related literature and industrial data are **10–15 years old**, despite rapid technological improvements. As a result, much of the published literature may not be representative of current operations.
8. Across all reviewed sources, the combined gaps in transparency, outdated values, and missing parameters highlight the need for **primary mill-level data collection**.

▼ 3. Life cycle inventory modelling

This chapter presents the life cycle inventory developed following the literature review and secondary data collection for weaving, taking into account the identified process variants.

3.1 LCI for sizing

For woven fabric production, sizing is an important step because it increases the efficiency of weaving operations.

3.1.1 Energy

Sizing is the only process step within weaving preparation that requires thermal energy. Indirect heating may be supplied through electricity, coal, or natural gas combustion via boilers.

Most studies indicate that the energy demand of sizing is driven mainly by the machine's fixed mechanical load, which remains largely independent of yarn count, fibre type (cotton vs. synthetics), or the specific sizing material used.

Research on textile processing consistently reports that electrical consumption for warp transport, immersion, squeezing, and cylinder rotation is effectively constant per machine at a given speed.

Therefore, assuming the same electrical energy inputs across yarn counts and fibre types is considered reasonable and is not expected to materially affect LCI accuracy.

Thermal energy consumption was estimated from the amount of water that must be evaporated during drying. This was calculated using the latent heat of evaporation, adjusted for typical dryer efficiency.

3.1.2 Size material

The life cycle inventory (LCI) for sizing material was developed by first defining representative natural and synthetic sizing formulations and converting them to a normalized basis per kilogram of yarn processed. The formulations were translated into total solids and water fractions to establish the sizing-liquor composition, from which the overall solids concentration was calculated. This concentration provides the basis for determining how much sizing material is deposited on the yarn.

Sizing demand was then linked to yarn characteristics by establishing a relationship between yarn fineness and size add-on percentage. Based on the selected yarn count, the corresponding dry add-on was estimated and converted into wet pick-up using the calculated solids concentration. This made it possible to determine both chemical uptake and water uptake per kilogram of yarn. A fibre-specific moisture-retention factor was applied to account for water that remains in the yarn after sizing, ensuring that only excess water is considered for evaporation in subsequent steps.

The LCI includes all key inputs, such as sizing chemicals (for example starch, binders, and lubricants) and process water. The detailed inventory is compiled and presented in the annex.

3.2 Life cycle inventory for weaving

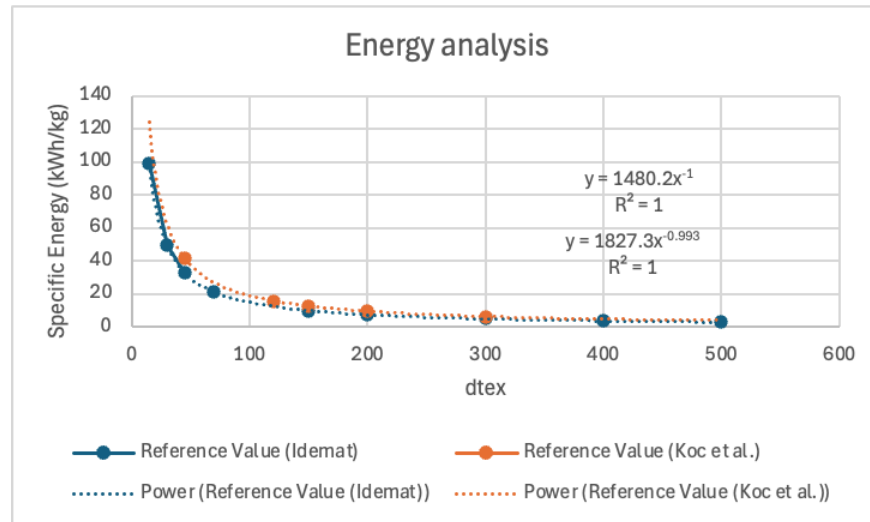
The following section presents the LCI for weaving, which includes the sizing operation as an input. Details of the sizing LCI are provided in the previous section.

3.2.1 Energy

The weaving energy model in this LCI covers the full set of preparatory and weaving operations—warping, sizing, drawing-in, loom operation, and inspection—as a combined energy value per kilogram of woven fabric.

Additionally, for transparency and diagnostic purposes, the underlying split of electricity across these individual subprocesses is retained in case root-cause analysis or sensitivity checks are required.

To model the effect of yarn dtex on weaving energy, all available literature reporting weaving-related energy consumption at known yarn fineness values, including [Koç et al. \(2010\)](#) and the Idemat database, was tabulated across multiple dtex values. For each source, a best-fit curve was developed, typically inverse-power or logarithmic, consistent with weaving theory, in which finer yarns require more picks per unit area.



The average of the best-fit curves was then evaluated and used to calculate weaving energy for all dtex values in the inventory.

$$Electricity(kWh/kg) = 1653.6(dtex)^{-0.996}$$

To account for variation across loom technologies, the model applies specific energy-consumption ratios between loom types (air-jet, rapier, projectile, shuttle, water-jet, and Jacquard), based on published comparison from [Prassana et al. \(2016\)](#).

3.2.2 Capital goods

As in knitting datasets, the building and machinery associated with 1 kg of weaving output were calculated using machinery weight, lifetime, and productivity (kg/h). Building infrastructure values were taken from the ecoinvent dataset for weaving. Additional information on capital-good assumptions is provided in the LCI file.

3.2.3 Yarn loss

Loom type is the dominant driver of yarn loss during weaving because selvedge method, weft insertion system, lubrication and oiling losses, moving-part complexity, and stoppage profile all vary strongly by loom. Yarn fineness (dtex/Ne) also matters because it affects splice loss and breakage probability, but its influence is secondary and is generally expressed through failure modes controlled by loom type and process settings.

Yarn losses are **modelled externally via the IOR**: the weaving dataset itself contains no waste-yarn exchange. The caller applies the loom-specific IOR to the upstream material input so that the yarn-production burden scales with the actual loss rate.

Different yarn-loss rates were assigned across loom types based on a review of available literature and industry observations. (Hague 2018), (Daffodil International University). The corresponding **input/output ratio (IOR)** expresses the mass of input yarn required per 1 kg of woven fabric:

$$IOR = \frac{1}{1 - \text{loss rate}}$$

Loom type	Yarn loss	IOR (kg input / kg output)
Water-jet	4%	1.041666667
Air-jet	6%	1.063829787
Projectile	6%	1.063829787
Rapier	7%	1.075268817
Jacquard	9%	1.098901099
Shuttle	12%	1.136363636

Water-jet looms were assumed to have the lowest losses (~4%), particularly for synthetic yarns, due to efficient insertion and reduced mechanical handling. Air-jet and projectile looms were assumed to have relatively lower losses (~6%) due to efficient weft insertion and fewer stoppages, while rapier looms were assigned slightly higher losses (~7%) because of increased selvage consumption and moderate breakage. Jacquard looms were assumed to have intermediate losses (~9%) because of additional mechanical complexity and auxiliary yarn consumption. In contrast, shuttle looms were assigned significantly higher losses (~12%), reflecting older technology with greater manual intervention, higher selvage waste, and more frequent stoppages.

▼ 4. Life cycle impact assessment

Impacts were computed using Brightway2 for climate change (kg CO₂eq).

4.1 LCIA results

Weaving route	GHG impact (kgCO ₂ e/kg)
weaving, 45 dtex-41 denier- 130/1 ne-222 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	22.2
weaving, 70 dtex-63 denier-84/1 ne-143 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	14.95
weaving, 120 dtex-108 denier-49/1 ne-83 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	9.43
weaving, 150 dtex-135 denier-40/1 ne-67 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	7.86
weaving, 170 dtex-153 denier-34/1 ne-59 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	7.11
weaving, 200 dtex-180 denier-30/1 ne-50 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	6.26
weaving, 250 dtex-225 denier-24/1 ne-40 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	5.27
weaving, 300 dtex-270 denier-20/1 ne-33 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	4.57
weaving, 330 dtex-297 denier-18/1 ne-30 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	4.25

weaving, 370 dtex-333 denier-16/1 ne-27 nm service, Fabric, Weaved production, at plant service, Fabric, Weaved	3.87
weaving (different fabric counts) service, Fabric, Weaved, Mix of different counts production mix, at plant service, Fabric, Weaved, Mix of different counts	8.58
weaving, air-jet service, Fabric, Weaved by air-jet loom production mix, at plant service, Fabric, Weaved by air-jet loom	10.99
weaving, generic water-jet service, Fabric, Weaved water-jet loom production mix, at plant service, Fabric, Weaved water-jet loom	3.88
weaving, jacquard service, Fabric, Weaved by jacquard loom production mix, at plant service, Fabric, Weaved by jacquard loom	9.82
weaving, rapier service, Fabric, Weaved by rapier loom production mix, at plant service, Fabric, Weaved by rapier loom	8.96
weaving, shuttle service, Fabric, Weaved by shuttle loom production mix, at plant service, Fabric, Weaved by shuttle loom	7.25
weaving, others service, Fabric, Weaved production mix, at plant service, Fabric, Weaved	8.18
weaving, water jet loom, 83 dtex- 75 den- 12 nm service, Fabric, Weaved water-jet loom, SpinDye® production, at plant service, Fabric, Weaved water-jet loom, SpinDye®	4.53
weaving, (150 dtex-135 denier-40/1 ne- 67 nm), cotton (twill fabric) service, Fabric, Weaved production, at plant service, Fabric, Weaved	8.24
weaving, silk service, Fabric, Silk production mix, at plant service, Fabric, Silk	11.08
weaving of carpet production, at plant Carpet, Weaved	6.09
weaving of primary backing service, Backing fabric, weaved production mix, at plant service, Backing fabric, weaved	1.65
sizing, natural fibers; single-end sizing; production mix, at plant; service (raw materials excluded)	0.775
sizing, synthetic fibers; single-end sizing; production mix, at plant; service (raw materials excluded)	0.643

4.2 Comparison with EF 3.1 datasets

Overall, the Carbonfact LCI differs from the EF 3.1 datasets through a higher level of process specificity and closer alignment with literature-based assumptions.

On average, Carbonfact shows approximately 10–11% lower electricity consumption, partly because of differences in electricity datasets and the use of yarn count-dependent sizing energy, whereas EF 3.1 applies a single averaged value without specifying yarn fineness or size composition.

In addition, EF 3.1 does not consistently apply literature-reported energy differences across loom types, most notably underestimating the lower energy demand of water-jet looms. Carbonfact, by contrast, aligns with published ratios, resulting in more realistic differentiation across technologies.

Finally, Carbonfact extends the system boundary by incorporating elements such as yarn loss, lubrication, and capital goods, which are not clearly accounted for in EF 3.1. This leads to a more comprehensive and transparent representation of weaving processes.

▼ 5. Inferences and limitations

Flexibility of the Carbonfact weaving LCI model

- The new LCI model allows weaving impacts to be calculated for any yarn count, enabling scenarios beyond the limited dtex ranges provided in EF datasets.

- Energy consumption in weaving can now be dynamically linked to yarn count, replacing EF's static, count-independent assumptions.
- Lubricating-oil use is included based on literature-backed values, providing a more complete process inventory than EF datasets.
- Yarn loss (%) can be varied by loom type and updated later when primary data becomes available, improving accuracy relative to EF's generic assumptions.
- The model supports optional inclusion or exclusion of the sizing process, allowing realistic modelling of fabrics and materials that do not undergo sizing (for example carpet, primary backing, and silk).

Sizing process enhancements

- The sizing module distinguishes between **natural and synthetic yarns**, enabling material-specific sizing impacts.
- Users can modify **size composition**, which affects size pick-up, water uptake, and water evaporation, allowing much higher process fidelity.
- If drying energy is known, it can be **directly entered** to compute thermal energy from evaporation, providing a traceable and transparent energy-modelling approach.

Data gaps and priorities

- To further strengthen model accuracy, **primary weaving data for silk, carpet, and primary backing fabrics** is needed, since EF provides energy values without process descriptions.
- **Water-jet weaving** remains a significant uncertainty: Carbonfact results differ from EF values by **more than 50%**, whereas all other loom types differ by only **5–20%**, indicating the need for reliable industry benchmarks.

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▼ 7. Annex

LCI for 1 kg of input yarn undergoing sizing (example: natural fibre and 197 dtex)

Activity	Value	Unit	Source / Rationale
Sizing agent	0.109664	kg	Derived from 8% of total size chemicals , as reported in the sizing supplier's technical data for synthetic yarn formulations. The value is scaled according to the total solids requirement calculated from yarn count-dependent size add-on.
Wax	0.009536	kg	Corresponds to 92% of total sizing auxiliaries based on the supplier's material composition breakdown. Wax contributes to lubrication and abrasion resistance and is included proportionally to the sizing solids applied.
Water	1.2004	kg	Computed from the wet pick-up derived from size add-on (%) and liquor solids concentration (%). Water requirement varies by yarn count and liquor composition; the detailed calculation procedure is described in the methodology section.
Building hall	6.90E-05	m ²	Allocated share of building area using the ecoinvent dataset : *textile production, cotton, weaving
Building machinery	1.17E-08	item	Estimated from the total weight and expected lifetime of sizing machinery. Infrastructure impacts are amortised over total production output to obtain use-phase allocation.
Electricity	0.264722222	kWh	Electricity consumption per kg of yarn sourced from the Resource

Activity	Value	Unit	Source / Rationale
			Efficient Management of Energy (REME) – SIA Project (Bangladesh) , which provides benchmark energy use for sizing, drying, and auxiliary operations.
Thermal energy	5.58186	MJ	Computed from the amount of water evaporated (difference between water pick-up and moisture retention), multiplied by the latent heat of evaporation and divided by the assumed dryer efficiency. The full methodology is presented in the main report.
Wastewater	0.0558186	kg	Assumes 5% loss of size liquor in the form of overflow, drainage, and preparation losses, consistent with typical sizing-room loss factors reported in technical guidelines.
Water to air	1.0605534	kg	Represents 95% of water evaporated in the dryer section, based on standard operating assumptions for slasher sizing where nearly all absorbed water is removed thermally.
End-of-life machinery	1.76E-05	kg	Calculated using the sizing-machine mass and service lifetime , with end-of-life impacts allocated proportionally to throughput.
Output: sized yarn	1.203228	kg	Total output mass equals the input dry-yarn mass plus retained moisture and applied size solids after drying.

Life cycle inventory of the weaving operation for 197 dtex natural fibre

Activity	Value	Unit	Source / Rationale
Lubricating oil	0.0305	kg	Value taken from Roos et al. (2015), Chemical LCI Inventory, which provides default lubricant consumption factors

Activity	Value	Unit	Source / Rationale
			for weaving machinery. This represents typical oil losses from central and point lubrication systems under standard operating conditions.
Sizing operation (sized warp input)	1.203	kg	Represents the input warp yarn after sizing, entering the weaving stage. The mass reflects size add-on and moisture retention calculated in the sizing LCI and is documented in the sizing methodology section.
Building hall	6.90E-05	m ²	Derived from the ecoinvent dataset: "textile production, cotton, weaving".
Building machinery	4.93E-07	item	Calculated from the total mass of weaving machinery (loom plus auxiliary devices) and its expected technical lifetime. The infrastructure burden is amortized over total loom output.
Electricity	8.57318	kWh	Calculated based on the yarn count-power relationship.
Waste yarn	Depending on IOR for loom type	kg	
Waste mineral oil	0.0305	kg	Estimated using ecoinvent waste flows for weaving and aligned with the assumed lubricating-oil input.
End-of-life machinery	2.22E-03	kg	Calculated through capital-goods amortization: machinery mass divided by total lifetime production. Includes end-of-life steel disposal burdens.
Output woven fabric (from sized yarn)	1.13103432	kg	Resulting mass of woven fabric after subtracting yarn-waste fractions from the sized-yarn input.